

Indian Institute of Technology Roorkee
Department of Applied Mathematics and Scientific Computing

Evaluating CNN and ViT on Skin Cancer Detection

A Summer Internship Report

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Certificate

This is to certify that the internship report entitled “Evaluating CNN and ViT on Skin Cancer Detection”, submitted by Khushi Jha, is a record of the work carried out by her during the Summer Internship Programme (25 May 2026 – 17 July 2026) at the Department of Applied Mathematics and Scientific Computing, Indian Institute of Technology Roorkee, under my supervision.

The work presented in this report, to the best of my knowledge, is an authentic account of the intern’s own study, implementation, and analysis, and has not been submitted elsewhere for the award of any other degree or diploma.

Place: Roorkee
Date: July 17, 2026
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Acknowledgement

Seven weeks ago, “Vision Transformer” was just a term I had seen in a paper title and quietly hoped nobody would ask me to explain. Writing that sentence now, at the end of this report, is a good way to measure how much this internship actually gave me.

I am deeply grateful to my supervisor, Prof. Millie Pant, for giving me the room to learn a fairly large field from first principles instead of jumping straight to results, and for the kind of feedback that pushes you to think rather than just to fix a plot.

I would like to sincerely thank my mentor, Mr. Shubham Joshi, for the many patient conversations about attention mechanisms, broken training runs, and the occasional confusion between a tensor’s shape and my own expectations of it. His guidance shaped not just what I built, but how I approached debugging, reading papers, and being honest about what did and did not work.

I also thank the Department of Applied Mathematics and Scientific Computing, IIT Roorkee, for hosting this internship and providing the computational resources without which none of the experiments in this report would exist.

Finally, to everyone who patiently listened to me explain the difference between a CNN and a ViT far more times than was strictly necessary – thank you for the patience.

Khushi Jha

Abstract

Skin cancer, and melanoma in particular, is one of the few cancers where the difference between an early diagnosis and a late one can be the difference between a minor procedure and a life-threatening illness. Yet the standard diagnostic tools available to a clinician – heuristics such as the ABCDE and CASH rules – still leave unaided visual inspection accuracy in the range of 60–80%. This gap is precisely where computer vision has spent the last decade trying to help, first with Convolutional Neural Networks (CNNs), and more recently with Vision Transformers (ViTs).

This report documents a seven-week summer internship focused on understanding, implementing, and critically evaluating this progression – from classical image processing to CNNs, to ViTs, to a hybrid CNN-Transformer multi-modal architecture. The internship began with the fundamentals of computer vision (edge detection, convolution, pooling), moved through transfer learning and the Vision Transformer architecture, and then applied both DenseNet121 and ViT-B/16 as baseline classifiers on the MILK10k (ISIC 2025) dataset. In parallel, a recent multi-modal explainable-AI framework from the literature, XAAI-Ledger, was reproduced from its published methodology on the ISIC 2019 dataset, including its Grad-CAM and SHAP based explainability components.

The results tell a fairly honest and, at times, humbling story: reproduced accuracy fell well short of the numbers reported in the source paper, largely due to class imbalance and the limited compute and time available within an internship timeline – a finding that is, in itself, a useful and common lesson in applied deep learning. The report closes by proposing a concept-bottleneck extension for future work, aimed at making the model’s diagnostic reasoning traceable to clinically meaningful concepts rather than an opaque probability score.

This document is organised as a standard internship report, with Chapter 1 introducing the problem and motivation, Chapter 2 covering the theoretical background required to understand every architecture used later, and the remaining chapters (methodology, experiments, results, and future work) building on this foundation.

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Chapter 1

Introduction

Every year, a huge number of skin lesions get looked at, squinted at, photographed, and sometimes biopsied – all to answer one deceptively simple question: is this dangerous, or not? Most of the time the answer is no. But the small fraction of times it is yes, and specifically when that “yes” is melanoma, timing matters more than almost anything else in oncology. This chapter sets up why that timing problem exists, why artificial intelligence has become a serious candidate for closing the gap, and what this internship specifically set out to do about it.

1.1 Artificial Intelligence in Healthcare

AI in healthcare did not begin with deep learning, even though it can feel that way from the inside of 2026. The earliest serious attempt at bringing “intelligence” into clinical decision-making was MYCIN in the 1970s – a rule-based expert system that reasoned about bacterial infections using a set of hand-coded if-then rules. It worked reasonably well on paper, and essentially never left the paper, because rule systems are brittle: every new edge case needs a new rule, and medicine is mostly edge cases.

The next real shift came with statistical machine learning in the 1990s and 2000s, particularly in radiology, where Computer-Aided Diagnosis (CAD) systems started flagging suspicious regions in mammograms and chest X-rays using hand-engineered features fed into classifiers like support vector machines. These systems were an improvement, but they were only ever as good as the features a human decided to extract – and deciding which features matter for a disease that presents in a thousand different ways is, itself, an extremely hard problem.

Deep learning changed the question entirely. Instead of an expert manually deciding that “asymmetry” or “edge irregularity” matters, a convolutional network could be shown thousands of labelled images and left to discover which patterns actually correlate with the diagnosis. The 2012 ImageNet moment (AlexNet) is usually cited as the spark for the broader field, and by 2016 the same idea had reached dermatology directly, with CNN-based classifiers matching board-certified dermatologists on curated benchmarks. Vision Transformers entered medical imaging a few years later, bringing a different inductive bias – global self-attention instead of local convolution – and the last few years have pushed toward multi-modal, explainable systems that try to combine image evidence with patient metadata and justify their own predictions. Figure 1.1 lays this progression out on a single timeline.

It is worth being honest about what CAD systems actually are before we go further, because the term gets thrown around loosely. A CAD system does not make a diagnosis. It makes a suggestion – a second opinion that a clinician is free to override. This distinction matters a lot for how the rest of this report should be read: none of the models discussed here are being proposed as replacements for a dermatologist. They are being evaluated as candidates for exactly this second-opinion role.

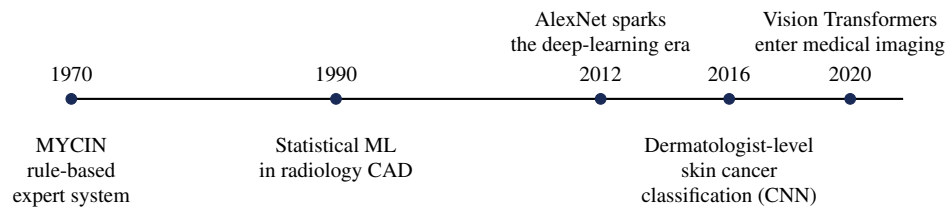


Figure 1.1: A brief evolution of AI in healthcare, from rule-based expert systems to modern multi-modal, explainable clinical AI.

1.2 Computer Vision in Medical Imaging

Medical image analysis has walked the same path that computer vision walked more broadly, just a little later and a little more cautiously (understandably – mistakes here have real consequences). That path looks roughly like Figure 1.2: it starts with traditional, hand-crafted image processing, moves into general deep learning, narrows into convolutional architectures built specifically for images, and most recently widens again into transformer-based architectures that were originally designed for language.

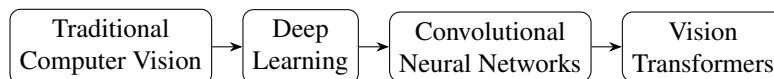


Figure 1.2: The broad evolution of computer vision techniques used in medical imaging, from hand-crafted filters to attention-based transformer architectures.

Traditional computer vision relied on filters designed by people – edge detectors, blob detectors, texture descriptors – which is precisely the kind of hand-engineering that made early CAD systems brittle. Deep learning replaced hand-designed features with learned ones, CNNs specialised that idea for grid-structured data like images, and Vision Transformers are the most recent attempt at asking whether a more general, attention-based architecture can do the same job while modelling relationships across the entire image rather than a local neighbourhood. Chapter 2 is dedicated entirely to unpacking every stage of this diagram in technical detail; this section exists only to give the reader a map before we start walking through it.

1.3 Skin Cancer Detection

Skin cancers broadly split into two families. Non-melanoma skin cancers – primarily basal cell carcinoma (BCC) and squamous cell carcinoma (SCC) – are common, usually slow-growing, and rarely fatal if treated. Melanoma is far rarer but disproportionately dangerous: it accounts for a small fraction of skin cancer cases but the large majority of skin-cancer deaths, because it can metastasize to other organs if left untreated.

This is exactly why early diagnosis is the single biggest lever available. Localised melanoma, caught before it spreads, has a five-year survival rate above 99%. Once it has spread to distant organs, that number falls below 30%. There is no equivalent jump anywhere else in the treatment pipeline – no drug, no surgical technique – that moves survival rates this much. The entire value proposition of an AI screening tool for skin cancer rests on this one fact: if it can nudge diagnosis earlier, even slightly, it is doing something that really matters.

Clinically, dermatologists use two well-known heuristics to structure their visual inspection, both referenced directly on the ISIC dataset documentation and in the wider dermatology literature:

- **ABCDE rule** – Asymmetry, Border irregularity, Color variation, Diameter (typically > 6 mm), and Evolution (change over time) [2].
- **CASH rule** – Color, Architecture, Symmetry, and Homogeneity, often used as a complementary framework in dermatoscopic assessment [3].

Figure 1.3 sketches the ABCDE criteria schematically – not as clinical photographs, but as a simplified visual mnemonic, which is really all the rule is meant to be in the first place.

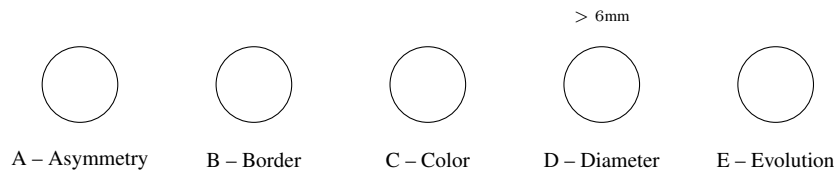


Figure 1.3: Schematic illustration of the ABCDE rule used in manual dermatological screening (Asymmetry, Border irregularity, Color variation, Diameter, Evolution). Illustration is a simplified mnemonic diagram, not a clinical image.

The catch, and the reason this whole report exists, is that these rules depend entirely on the skill and consistency of the person applying them. Two different clinicians looking at the same lesion do not always agree, and inter-observer variability is a well-documented problem in dermatology, with even experienced clinicians’ unaided accuracy sitting around 60–80% [1]. This is the gap AI is being asked to fill: not to replace clinical judgement, but to offer a second, consistent opinion that does not get tired, does not have an off day, and has effectively seen more lesions than any single clinician ever will.

1.4 Problem Statement

Framed plainly: manual visual screening for melanoma is inconsistent, heuristic-based, and caps out at around 60–80% accuracy even among experienced clinicians [1]. Deep learning models promise a more consistent alternative, but in practice they inherit a set of problems that are easy to state and surprisingly hard to fix:

- **Visual similarity across classes.** Early melanoma and a benign nevus can look almost identical to the naked eye (and, it turns out, to a CNN too) – the differences are often subtle textural cues rather than obvious shape differences.
- **Class imbalance.** Malignant cases are, thankfully, rare in the population – which is great for public health and genuinely unhelpful for training a classifier. Figure 1.4 shows exactly how lopsided this gets on the ISIC 2019 dataset used later in this report: benign cases outnumber melanoma cases by more than four to one.
- **Limited and inconsistent data.** Dermoscopic datasets vary in imaging device, lighting, patient demographics, and label quality, which makes a model trained on one dataset fragile when it meets another.
- **Metadata ignorance.** A large share of published skin-cancer classifiers are image-only, even though a dermatologist would never ignore a patient’s age, sex, or the anatomical site of a lesion when forming a judgement.
- **Lack of explainability.** A model that outputs “87% melanoma” with no justification is not something a clinician can act on with confidence – and regulatory approval for clinical AI increasingly requires some form of interpretability.

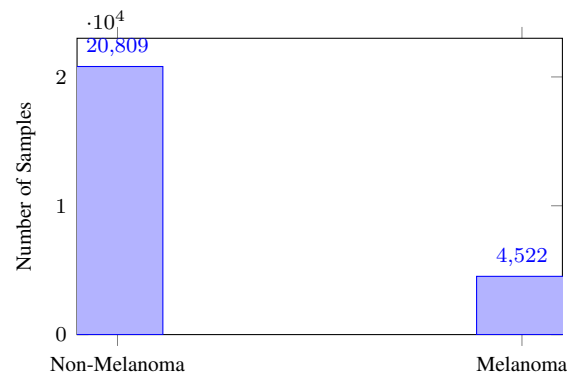


Figure 1.4: Class imbalance in the ISIC 2019 dataset used in this internship: 20,809 non-melanoma samples versus 4,522 melanoma samples – a ratio of roughly 4.6:1.

Every one of these five problems reappears later in this report, either as a design consideration in the models that were built, or as a direct explanation for why the reproduced results in later chapters fall short of the numbers reported in the literature.

1.5 Motivation

It would be easy to say “CNNs work well for images, so let’s use a CNN,” and honestly, that was more or less my starting assumption before this internship. A few weeks in, that assumption did not survive contact with the literature or the data.

CNNs are excellent at picking up local texture – edges, colour gradients, small-scale patterns – because that is exactly what a convolutional filter is built to do. But melanoma diagnosis often depends on relationships across the whole lesion: is the colour distribution consistent from one edge to the other, does the border look different on one side than the other. A convolution kernel, by design, only looks at a small neighbourhood at a time; it has to stack many layers before it can reason about the image as a whole, and even then, that global reasoning is implicit and diluted.

Vision Transformers were the obvious next thing to try, because self-attention lets every patch of the image attend to every other patch from the very first layer – global context, right from the start. But ViTs are notoriously data-hungry, and the datasets available in dermatology, while not tiny, are nowhere close to the scale ViTs were originally designed for. So neither architecture, taken alone, is obviously “enough.”

The piece that convinced me the field is moving in the right direction, though, was metadata. A 45-year-old patient with a changing lesion on sun-exposed skin is a different clinical picture from a 20-year-old with a stable lesion on a rarely-exposed site – even if the two lesions look nearly identical in a photograph. Image-only models are, by construction, blind to this information. That is the real motivation behind this internship: not just to benchmark CNN against ViT, but to understand why the field is increasingly moving toward architectures that fuse visual features with structured clinical metadata, and to reproduce one such architecture end-to-end to see, honestly, how well it holds up outside its original paper.

1.6 Objectives of the Internship

Rather than framing this as a research contribution with novel objectives, it is more honest to frame it the way it actually happened – as a structured learning path with concrete checkpoints:

1. Build a solid working understanding of classical and deep-learning-based computer vision, starting from edge detection up to convolutional networks.
2. Understand why CNNs work the way they do – not just how to call `Conv2D` in a framework, but what the convolution and pooling operations are actually doing to an image.
3. Study the Vision Transformer architecture in enough depth to explain patch embedding, positional encoding, and multi-head self-attention without reaching for a slide deck.
4. Implement two independent baseline classifiers – DenseNet121 and ViT-B/16 – on a modern multi-class dermatology dataset (MILK10k), and compare them fairly under identical training conditions.
5. Reproduce a recent published multi-modal, explainable architecture (XAAI-Ledger) from its methodology section alone, including its Grad-CAM and SHAP explainability components.
6. Explore what a meaningful next step beyond this architecture could look like, grounded in the specific weaknesses observed during reproduction.

1.7 Internship Timeline

The seven weeks of the internship were structured to build up complexity gradually – theory before implementation, single-modality baselines before multi-modal fusion, and reproduction before proposing anything new. Table 1.1 and Figure 1.5 summarise this plan.

Table 1.1: Week-by-week internship work plan.

Week	Work Done
Week 1	Computer Vision Basics
Week 2	Vision Transformer
Week 3	DenseNet Baseline
Week 4	ViT Baseline
Week 5	XAAI-Ledger Implementation
Week 6	Proposed Improvements
Week 7	Analysis

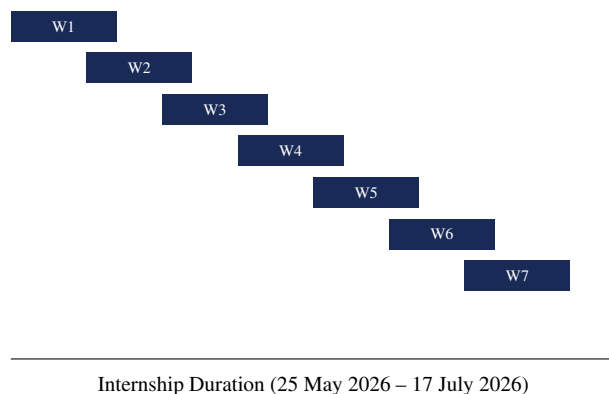


Figure 1.5: Seven-week internship timeline, from foundational computer vision concepts through to final analysis.

The chapters that follow roughly mirror this timeline. Chapter 2 covers everything learned in Weeks 1–2; the baseline implementations, the XAAI-Ledger reproduction, and the proposed improvements and analysis from the later weeks are covered in the chapters that follow it.

Chapter 2

Background Study

This chapter is, essentially, seven weeks of notes cleaned up into something readable. It has nothing to do with skin cancer specifically – that connection is made later. What it covers is the computer vision toolbox itself: how images were processed before deep learning existed, why convolutional networks became the default choice once deep learning arrived, and how Vision Transformers arrived to challenge that default. By the end of this chapter, every architecture used later in this report should feel like a natural extension of something explained here, rather than a black box pulled off a shelf.

2.1 Traditional Computer Vision

Before a neural network ever learns a single feature on its own, it is worth understanding what a feature even is, because traditional computer vision made this completely explicit. An image, at its core, is just a grid of numbers – pixel intensities. Traditional CV techniques are, almost without exception, small mathematical operations applied to this grid to pull out something structurally meaningful: edges, corners, blobs, textures.

2.1.1 Edge Detection

An edge is a place in the image where intensity changes sharply – the boundary of an object, say, or the border of a lesion. Mathematically, an edge corresponds to a large gradient in the image intensity function $I(x, y)$. Three filters come up again and again:

Sobel operator. Approximates the image gradient in the x and y directions using two small 3×3 kernels:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad (2.1)$$

The overall edge strength (gradient magnitude) at each pixel is then

$$|G| = \sqrt{G_x^2 + G_y^2}. \quad (2.2)$$

Gaussian blur. Before edge detection, images are usually smoothed with a Gaussian kernel to suppress noise, since raw gradients are extremely sensitive to it:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}. \quad (2.3)$$

Laplacian operator. Rather than a first-order gradient, the Laplacian is a second-order derivative operator, useful for detecting regions of rapid intensity change in all directions at once:

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}. \quad (2.4)$$

Figure 2.1 shows all three of these operators applied to the same test image, computed directly for this report rather than borrowed from a paper, so the reader can see exactly what each filter does to the same input.

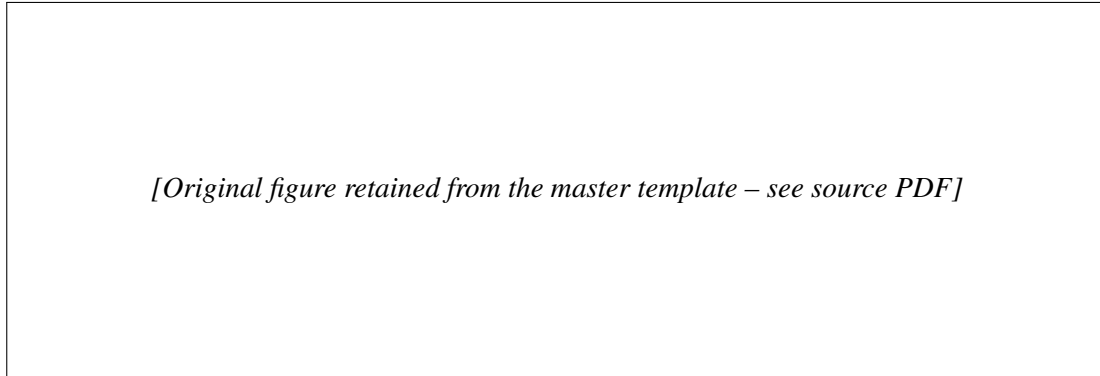


Figure 2.1: Gaussian smoothing, Sobel edge magnitude, and Laplacian filtering applied to the same grayscale test image, computed for this report.

The honest limitation of this entire family of methods is that a human had to decide, in advance, which filter matters. Sobel is good at edges. It is not good at detecting a subtle texture change that does not manifest as a sharp gradient. This is exactly the ceiling that deep learning eventually broke through – by learning the filters instead of hand-designing them.

2.2 Deep Learning Basics

An **Artificial Neural Network (ANN)** is, structurally, a sequence of layers, where each layer computes a weighted sum of its inputs followed by a non-linear activation function:

$$a^{(l)} = \sigma \left(W^{(l)} a^{(l-1)} + b^{(l)} \right), \quad (2.5)$$

where $W^{(l)}$ and $b^{(l)}$ are the learnable weights and bias of layer l , and σ is a non-linearity such as ReLU. Stack enough of these layers and, in principle, the network can approximate extremely complex functions – this is the universal approximation intuition behind deep learning.

The problem with applying a plain ANN directly to images is scale. A modestly sized 224×224 RGB image has $224 \times 224 \times 3 = 150,528$ input values. A fully-connected first layer with even 1,000 neurons would need over 150 million weights just for that one layer – computationally wasteful, and worse, it throws away the fact that nearby pixels are related to each other. A Convolutional Neural Network (CNN) fixes both problems at once, by replacing full connectivity with small, shared filters that slide across the image (the same idea as the Sobel filter above, except now the filter's numbers are learned rather than hand-picked). This gives CNNs two properties that make them well suited to images specifically:

- **Local connectivity** – each neuron only looks at a small patch of the image, which matches the intuition that a pixel is most related to its neighbours.

- **Weight sharing** – the same filter is reused across the entire image, drastically reducing the number of parameters and letting the network detect the same pattern (e.g. an edge) regardless of where it appears.

2.3 Convolution Operation

The convolution operation itself is simple enough to compute by hand for a small example, which is usually the fastest way to actually understand it. For a single-channel input I and a kernel K of size $k \times k$, the output at position (i, j) is

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i + m, j + n) K(m, n). \quad (2.6)$$

Three extra parameters control how this operation behaves in practice, and all three matter a great deal for the architectures used later in this report:

- **Kernel size** – the spatial extent of the filter (commonly 3×3 or 5×5). Larger kernels see more context per step but cost more computation.
- **Stride** – how far the kernel moves between applications. A stride of 2 roughly halves the output’s spatial dimensions.
- **Padding** – whether the input is padded with zeros at the border, so the kernel can be centred on edge pixels too (“same” padding keeps the output size equal to the input size; “valid” padding shrinks it).

For an input of size $W \times W$, kernel size F , stride S , and padding P , the output spatial size is

$$W_{out} = \left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1. \quad (2.7)$$

Figure 2.2 illustrates a 3×3 kernel sliding over a padded input with stride 1, producing one output feature map.

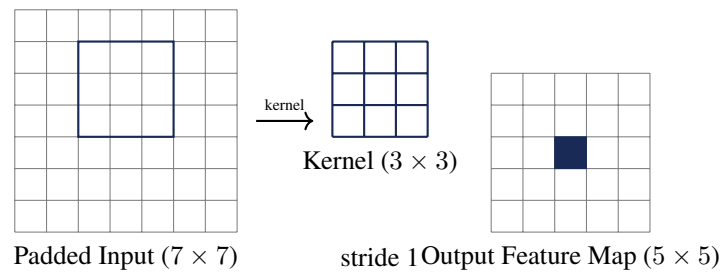


Figure 2.2: A 3×3 kernel sliding with stride 1 over a zero-padded input, producing a single output feature map. The highlighted receptive field on the input maps to a single highlighted output pixel.

Pooling. Convolutional layers are almost always followed by a pooling operation, which reduces the spatial resolution of the feature map while keeping the most important information. Max pooling, the most common choice, simply takes the maximum value within each small window:

$$P(i, j) = \max_{(m, n) \in \text{window}} S(i + m, j + n). \quad (2.8)$$

Pooling gives the network a small amount of translation invariance (a feature is still detected even if it shifts slightly) and reduces the parameter count of subsequent layers.

A typical CNN, then, is simply a repeated stack of Convolution → Activation → Pooling blocks, with the spatial size shrinking and the number of channels (feature types) growing at each stage, before a final classification head flattens everything into class probabilities.

2.4 Transfer Learning

Training a CNN or a ViT completely from scratch requires a very large labelled dataset – ImageNet, the standard benchmark, contains over a million labelled images across a thousand categories. Dermatology datasets, by comparison, are small. Transfer learning sidesteps this problem with a simple observation: the early layers of a network trained on natural images learn very general features (edges, colours, simple textures) that are useful for almost any visual task, not just the one the network was originally trained on.

The typical recipe is:

1. **Pretraining** – train a large CNN or ViT backbone on a large, general dataset (ImageNet).
2. **Feature extraction** – freeze the pretrained backbone’s weights and use it purely to convert an image into a feature vector.
3. **Fine-tuning** – optionally, unfreeze some or all of the backbone and continue training on the target dataset (here, dermoscopic images) at a low learning rate, so the general features adapt slightly to the new domain without being destroyed.

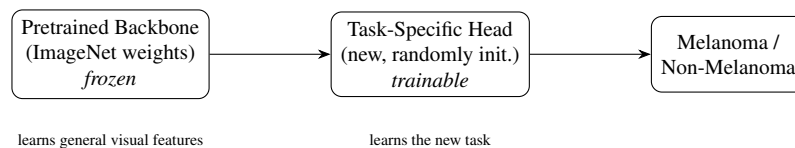


Figure 2.3: The transfer-learning recipe used throughout this report: a frozen, ImageNet-pretrained backbone (CNN or ViT) acts as a general-purpose feature extractor, feeding into a small trainable head adapted to the skin-cancer classification task.

Every baseline model built during this internship – DenseNet121, ViT-B/16, and the EfficientNetB0 backbone used inside the XAAI-Ledger reproduction – follows exactly this pattern, and this is the reason it is possible to get reasonable performance on a dataset with only tens of thousands of images rather than a million.

2.5 Vision Transformer

The Vision Transformer (ViT) takes the transformer architecture, originally built for language, and applies it to images with a surprisingly small number of image-specific changes [4]. The core idea: treat an image as a sequence of patches, exactly the way a sentence is a sequence of words, and let self-attention handle the rest.

2.5.1 Patch and Position Embedding

An input image of size $H \times W \times C$ is split into N non-overlapping patches of size $P \times P$, where $N = \frac{HW}{P^2}$. Each flattened patch is linearly projected into a D -dimensional embedding:

$$z_0 = [x_{class}; x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{pos}, \quad (2.9)$$

where E is the learnable patch-embedding matrix, x_{class} is a special learnable CLS token prepended to the sequence (its final-layer representation is later used for classification), and E_{pos} is a learnable positional embedding – necessary because, unlike a CNN, self-attention has no inherent notion of spatial order and needs to be told where each patch came from.

2.5.2 Transformer Encoder

The embedded sequence passes through L identical encoder blocks, each built from two sub-layers wrapped in residual connections and Layer Normalization:

$$z'_\ell = \text{MSA}(\text{LN}(z_{\ell-1})) + z_{\ell-1} \quad (2.10)$$

$$z_\ell = \text{MLP}(\text{LN}(z'_\ell)) + z'_\ell \quad (2.11)$$

where MSA is Multi-Head Self-Attention and MLP is a small feed-forward network (typically two linear layers with a GELU activation in between). The residual connections are what make it possible to stack many such blocks without gradients vanishing, and LayerNorm keeps the activations at each layer well-scaled during training.

2.5.3 Multi-Head Self-Attention

Self-attention is the mechanism that lets every patch look at every other patch. For a single attention head, each input token is projected into a Query, Key, and Value vector, and attention weights are computed as a scaled dot-product:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad (2.12)$$

where d_k is the dimensionality of the key vectors (the scaling factor prevents the dot products from growing too large and saturating the softmax). Multi-head attention simply runs h of these attention operations in parallel with different learned projections, and concatenates the results – allowing the model to attend to different types of relationships (e.g. colour similarity in one head, spatial proximity in another) simultaneously.

The upshot of all this machinery is that, from the very first encoder block, the CLS token can attend to every patch in the image directly. This is a fundamentally different inductive bias from a CNN, where a neuron in an early layer can only see a small local neighbourhood and has to wait for many layers of convolution before its receptive field covers the whole image.

2.6 CNN vs ViT: A Theoretical Comparison

With both architectures now unpacked, it is worth putting them side by side before any experiment is run. Table 2.1 summarises the theoretical trade-offs; the empirical side of this comparison, on an actual dermatology dataset, is picked up again later in this report.

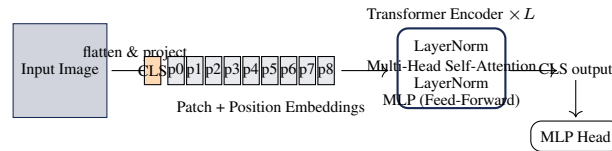


Figure 2.4: The Vision Transformer (ViT) pipeline: an image is split into patches, linearly embedded with a prepended CLS token and positional embeddings, passed through L transformer encoder blocks (each with multi-head self-attention, residual connections, and an MLP sub-layer), and the final CLS token representation is classified by a small MLP head.

Neither column in Table 2.1 is strictly better than the other – and that, ultimately, is the whole justification for the multi-modal, hybrid approach explored in the rest of this report. A CNN backbone is well suited to extracting fine local texture from a dermoscopic image; a transformer-based fusion layer is well suited to relating that visual evidence to structured clinical metadata as a global reasoning step on top. Chapters that follow build directly on both halves of this table – first evaluating DenseNet121 (a pure CNN) against ViT-B/16 (a pure transformer) as independent baselines, and then examining an architecture that tries to combine the strengths of both.

Table 2.1: Theoretical comparison between Convolutional Neural Networks and Vision Transformers.

Aspect	CNN	ViT
Inductive bias	Strong: locality, weight sharing, translation invariance	Weak: minimal built-in assumptions about image structure
Receptive field	Local at early layers; grows gradually with depth	Global from the very first layer, via self-attention
Data requirement	Performs well on small-to-medium datasets	Typically needs large-scale pretraining data to match CNN performance
Parameter efficiency	Fewer parameters for a given receptive field, due to weight sharing	Generally more parameters; self-attention is quadratic in sequence length
Interpretability tools	Grad-CAM, saliency maps (well established)	Attention-map visualisation (more direct, since attention weights are inherently interpretable)
Typical strength	Fine, local texture and edge patterns	Long-range dependencies and global context
Computational cost	Convolution is linear in image size	Self-attention scales quadratically with number of patches

Chapter 3

Literature Review

Chapter 2 built the toolbox – convolution, transfer learning, self-attention. This chapter puts that toolbox in context by asking a narrower question: what has actually been tried, by other people, for this specific problem, and what did they find? The chapter moves from general CNN-based skin cancer work through Vision Transformers, into explainability and multimodal fusion, and closes with a detailed review of the one paper this internship set out to reproduce – XAAI-Ledger – before stating, explicitly, the gap that motivates the implementation chapters that follow.

3.1 Deep Learning in Skin Cancer Detection

The shift from hand-crafted features to learned ones, introduced in general terms in Section 2.1, reached dermatology specifically once large public datasets became available. Romero-Lopez et al. [2] were among the early groups to show that a CNN trained directly on dermoscopic images could outperform classical feature-plus-SVM pipelines on lesion classification, without any hand-engineered ABCDE-style features at all. This result mattered less for its raw accuracy number, which has since been surpassed many times over, and more for what it demonstrated in principle: that a network could discover diagnostically relevant patterns – asymmetry, border irregularity, colour variation – on its own, directly from pixels, essentially rediscovering the ABCDE heuristic from data rather than being told it.

What followed this initial demonstration was less a single breakthrough than a fairly steady widening of scope – larger and more diverse datasets, deeper architectures, and, eventually, an entirely different way of processing an image altogether (Vision Transformers, discussed in Section 3.3). The rest of this chapter follows that widening in roughly the order it happened.

3.2 CNN-Based Skin Cancer Classification

Once CNNs were established as viable, the natural next question was which CNN. DenseNet [5], used later in this report as Baseline 1, connects every layer to every subsequent layer in a feed-forward fashion, which encourages feature reuse and, in practice, tends to need fewer parameters than an equally deep architecture without such connections for a comparable level of accuracy. This property is particularly attractive in dermatology, where labelled data is comparatively scarce and a smaller, well-regularised network is less prone to overfitting than a very large one.

A separate line of work focused on combining textural and colour-based features with an ensemble of CNN backbones rather than relying on a single network. Sharafudeen and Chandra [3] proposed a CASH-rule-based architecture that fuses texture features with a deep backbone specifically to keep

the model’s reasoning aligned with a criterion a dermatologist would already recognise, and a related ensemble-based follow-up [11] combined multiple CNN backbones with textural descriptors for improved robustness on abnormality detection. The consistent finding across this body of work is that CNN accuracy on curated, single-source benchmarks (such as HAM10000) is high – frequently exceeding 85–90% – but drops noticeably when the same model is evaluated on a dataset collected with a different imaging device or patient population, a generalisation gap that reappears directly in the results of this report (Chapter 5).

Table 3.1 summarises the CNN-based approaches discussed above.

Table 3.1: Comparison of representative CNN-based approaches to skin lesion classification.

Study	Backbone	Dataset	Reported strength	Reported limitation
Romero-Lopez et al. [2]	Custom CNN	Dermoscopic subset	First demonstration of learned-feature superiority over hand-crafted features	Small dataset, single-source
Huang et al. [5] (DenseNet)	DenseNet-121/169/201	ImageNet transfer	Parameter-efficient, strong feature reuse via dense connectivity	Not dermatology-specific; needs fine-tuning
Sharafudeen and Chandra [3, 11]	CASH-rule + texture fusion	Curated dermoscopic set	Keeps model reasoning aligned with a clinical heuristic	Ensemble adds inference cost and complexity

3.3 Vision Transformer-Based Classification

Once ViTs [4] demonstrated competitive image classification performance without any convolutional inductive bias, dermatology researchers began asking whether the same held for skin lesions specifically. The general finding, echoed across several follow-up studies, is a variant of the trade-off already summarised theoretically in Table 2.1: a ViT trained from scratch on a dermatology-sized dataset underperforms a CNN of comparable size, but a ViT that starts from an ImageNet-pretrained checkpoint and is fine-tuned – exactly the transfer-learning recipe of Section 2.4 – closes most or all of that gap, and in several published comparisons slightly exceeds the CNN baseline on global, texture-independent cues such as overall lesion shape and colour distribution.

Hybrid CNN-transformer encoders, first popularised outside dermatology by architectures such as TransUNet [9] for medical image segmentation, extended this idea further by using a CNN as a feature extractor and a transformer as the layer that reasons globally over those features, rather than choosing one architecture family exclusively. This “CNN-for-locality, transformer-for-globality” pattern recurs throughout the multimodal literature reviewed in Section 3.5, and is the direct conceptual ancestor of the fusion architecture reproduced in Chapter 6.

3.4 Explainable AI in Medical Imaging

A classifier that only outputs a probability is difficult to act on clinically, a point already raised in Section 1.4. Two explainability techniques dominate the dermatology literature and are used directly in this report’s own reproduction work (Chapter 6): Grad-CAM [8] and SHAP.

Grad-CAM produces a coarse localisation heatmap by using the gradients flowing into the final convolutional layer to weight that layer’s feature maps, highlighting the image regions that most influenced a given prediction. It is well suited to CNN-style architectures precisely because it needs a spatial con-

volitional feature map to operate on, and it has become close to a default requirement in published dermatology CNN papers as a sanity check that the network is attending to the lesion rather than to an imaging artefact (a ruler, a hair, a border marking pen).

SHAP takes a complementary, feature-level approach based on Shapley values from cooperative game theory: it estimates each input feature’s marginal contribution to the final prediction. For image-only models SHAP is less commonly used than Grad-CAM, but for the tabular metadata branch of a multimodal model – age, sex, anatomical site – it is close to the standard tool, since Grad-CAM has no equivalent for a non-spatial input. Table 3.2 summarises the two techniques side by side.

Table 3.2: Grad-CAM versus SHAP as explainability tools in this report.

Aspect	Grad-CAM	SHAP
Input type it explains	Spatial (image)	Tabular / structured
Mechanism	Gradient-weighted class activation	Shapley value attribution
Output	Pixel-level heatmap	Per-feature contribution score
Used on	CNN image branch	Metadata (age, sex, site) branch

Neither technique explains why the model considers a concept diagnostically relevant in the first place – both operate after the fact, on a model that was never asked to represent clinical concepts explicitly. This limitation is picked up directly in Section 3.7 and again in the future-work discussion of Chapter 7.

3.5 Multimodal Learning for Skin Cancer

The metadata-ignorance problem raised in Section 1.4 is precisely what multimodal architectures are designed to address. The general pattern across published multimodal skin-cancer work is a two-stream design: an image branch (typically a pretrained CNN or ViT) produces a visual embedding, a separate branch processes structured metadata (age, sex, anatomical site, sometimes patient history), and a fusion mechanism – concatenation, an MLP, or, increasingly, cross-attention – combines the two before a final classification head.

A separate and, in recent years, empirically dominant family of multimodal approaches avoids end-to-end fusion entirely and instead stacks a CNN’s out-of-fold prediction probabilities with a gradient-boosted decision tree (GBDT) trained on tabular metadata. This two-stage “CNN + GBDT” pattern traces back to the winning solution of the SIIM-ISIC 2020 melanoma classification challenge (Ha, Liu, and Liu [12]) and was the dominant pattern again among top entries to the ISIC 2024 “Skin Cancer Detection with 3D-TBP” challenge, which typically ensembled multiple CNN/ViT backbones with several GBDT variants (LightGBM, XGBoost, CatBoost) across many cross-validation folds. Baseline 3 in this report (Section 5.4) is a deliberately scaled-down single-backbone, single-GBDT reproduction of exactly this recipe, and its results are read directly against this literature in that section.

Across both fusion styles, the consistent published finding is that metadata helps – but only when it is genuinely complementary to the image, and only when there is enough of it. A handful of coarse fields (age, sex, one anatomical-site category) tend to add a small, sometimes negligible improvement over an image-only baseline; richer metadata (patient history, multiple lesion sites over time, dermatologist-recorded features) adds substantially more. This distinction turns out to matter directly for the results in this report, where the metadata available is limited to exactly the coarse end of that spectrum.

3.6 Review of XAAI-Ledger

The specific paper this internship set out to reproduce, referred to throughout this report as XAAI-Ledger, sits squarely within the end-to-end fusion family described above. Its architecture, discussed in full technical detail in Section 6.2, couples a frozen pretrained CNN image branch with a small MLP metadata branch, projects both into a shared representation, and fuses them through a multi-head self-attention block before a final sigmoid classification head for binary melanoma versus non-melanoma prediction. The paper reports Grad-CAM and SHAP explainability outputs as a core part of its contribution, framing the work as much around transparency as around raw accuracy.

The paper’s headline results, reported on the ISIC 2019 dataset, are notably strong – accuracy and sensitivity figures in the high nineties, discussed in full in Table 5.7 and in Chapter 6 alongside this internship’s own reproduction of the same pipeline. What the published methodology section does not fully specify, and what turns out to matter a great deal in practice, is the exact class-imbalance handling strategy used during training; ISIC 2019 has a roughly 4.6:1 imbalance between non-melanoma and melanoma cases (Figure 1.4), and how that imbalance is compensated for – class weighting, oversampling, a custom loss – has a large effect on sensitivity specifically. Chapter 6 returns to this gap directly, since it is the leading candidate explanation for the difference between the paper’s reported numbers and the numbers reproduced in this internship.

3.7 Research Gap

Pulling the threads of this chapter together, three gaps stand out as directly relevant to the work that follows:

- **Benchmark comparability.** Published CNN and ViT results for skin lesion classification are rarely evaluated under an identical protocol – same dataset, same split, same number of epochs, same evaluation metrics – which makes head-to-head comparisons across papers unreliable. Chapters 4 and 5 of this report address this directly by training DenseNet121 and ViT-B/16 under exactly matched conditions on the same dataset.
- **Reproducibility of multimodal fusion claims.** Multimodal architectures, including XAAI-Ledger, are frequently proposed and evaluated only by their original authors, with implementation details – imputation strategy, class-imbalance handling, exact fusion hyperparameters – often under-specified in the published methodology. Chapter 6 reproduces XAAI-Ledger independently from its methodology section and reports the reproduced numbers honestly, imbalance-related shortfall included.
- **Interpretability that stops at attribution.** Grad-CAM and SHAP, reviewed in Section 3.4, explain a trained model’s existing decision after the fact; neither forces the model to reason through clinically meaningful concepts in the first place. This is the specific gap the Concept Bottleneck extension proposed in Chapter 7 is aimed at.

These three gaps motivate the structure of the rest of this report directly: Chapters 4 and 5 build and fairly compare the two single-modality baselines, Chapter 6 reproduces and honestly evaluates the multimodal reference architecture, and Chapter 7 proposes a concrete next step grounded in the third gap above.

Chapter 4

Dataset and Experimental Setup

Chapter 3 closed by identifying a comparability gap between published CNN and ViT results. This chapter describes exactly how that gap was closed within this internship – the datasets used, how they were prepared, and the precise experimental conditions under which every model in Chapter 5 was trained – so that the results reported there are fully reproducible from this description alone.

4.1 Experimental Workflow

All experiments in this report follow the same four-stage pipeline: dataset acquisition and cleaning, preprocessing, model training under a fixed protocol, and evaluation against a common set of metrics. Figure 4.1 lays this pipeline out end to end; the sections that follow expand each stage.

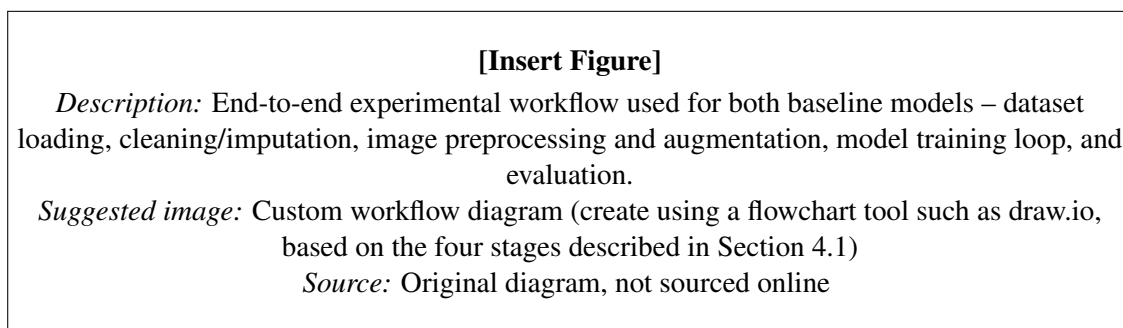


Figure 4.1: End-to-end experimental workflow used in this report.

4.2 Dataset Description

Three public dermatology datasets were used across this internship, each serving a distinct role. Table 4.1 summarises all three, alongside ISIC 2020, which was reviewed during dataset selection but not used for training.

MILK10k was selected as the primary baseline dataset because it is the most recent public release with genuine dual-image (clinical close-up plus dermoscopic) multimodality and explicit skin-tone annotation, making it the natural target for the concept-bottleneck extension proposed in Chapter 7. ISIC 2019 was retained specifically for the XAAI-Ledger reproduction, since it is the dataset used in the original paper.

Table 4.1: Datasets used or evaluated during this internship.

Dataset (Year)	Images	Classes	Metadata	Class balance	Role in this report
ISIC 2019	25,331	8 (MEL, NV, BCC, AK, BKL, DF, VASC, SCC)	Age, sex, anatomic site	17.9% melanoma (4,522 / 25,331)	Training/validation set for the XAAI-Ledger reproduction (Chapter 6)
HAM10000	10,015	7 (akiec, bcc, bkl, df, mel, nv, vasc)	Age, sex, lesion localisation	11.1% melanoma (1,113 / 10,015)	Cited directly in XAAI-Ledger; reviewed for cross-dataset context
MILK10k (ISIC 2025)	10,480 (5,240 paired lesions)	11 broad categories (up to 48 fine-grained ISIC-DX diagnoses)	Age, sex, anatomic site, skin tone (6-level), histopathology flag	95.7% histopathology-confirmed ground truth	Training/validation set for DenseNet121 and ViT-B/16 baselines (Chapter 5)

4.3 Dataset Analysis

MILK10k’s 11 diagnostic categories are not evenly represented; several rare classes contain fewer than 20 samples in the training split, which reappears as a direct explanation for near-zero per-class F1 scores in Section 5.4. ISIC 2019 has a coarser, 8-class label set and the roughly 4.6:1 non-melanoma-to-melanoma imbalance already illustrated in Figure 1.4.

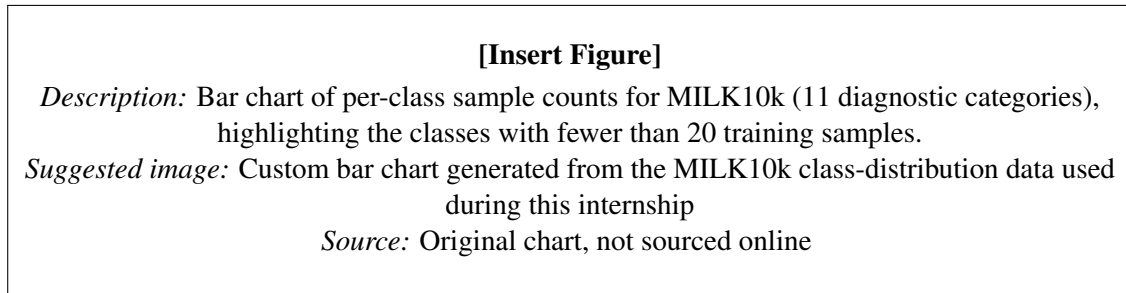


Figure 4.2: MILK10k per-class sample distribution.

4.4 Data Preprocessing

Both datasets went through a common preprocessing pipeline before being passed to a model:

- **Image resizing.** All dermoscopic images were resized to $224 \times 224 \times 3$ to match the input resolution expected by the ImageNet-pretrained DenseNet121, ViT-B/16, and EfficientNetB0 backbones used throughout this report.
- **Normalisation.** Pixel values were rescaled and normalised using the same per-channel mean and standard deviation each backbone was originally pretrained with, so that the pretrained low-level filters (Section 2.4) remain well matched to the input distribution.
- **Metadata imputation.** Missing tabular fields were filled using median imputation for numeric fields (age) and mode imputation for categorical fields (sex, anatomical site) – the same strategy used in the XAAI-Ledger reproduction (Section 6.3), applied here for consistency across the report.
- **Metadata encoding.** Categorical metadata fields were one-hot encoded, and continuous fields were scaled with a MinMaxScaler, prior to being concatenated or fed into any metadata-consuming branch (Baseline 3, Section 5.4).
- **Train/validation split.** A stratified split was used throughout, preserving class proportions in both partitions – 75/25 for DenseNet121 and ViT-B/16 on MILK10k, and 80/20 for Baseline 3 on MILK10k (Section 5.4).

Table 4.2 summarises the resulting split sizes.

Table 4.2: Train/validation split sizes used across baseline models.

Model	Dataset	Split ratio	Validation set size
DenseNet121	MILK10k	75/25	2,620
ViT-B/16	MILK10k	75/25	2,620
Baseline 3 (CNN + GBDT)	MILK10k	80/20	2,096
XAAI-Ledger reproduction	ISIC 2019	75/25 (stratified)	6,333 (5,202 non-melanoma, 1,131 melanoma)

4.5 Experimental Environment

All models were trained using the computational resources provided by the Department of Applied Mathematics and Scientific Computing, IIT Roorkee, using standard deep-learning frameworks with GPU acceleration.

Table 4.3: Experimental environment.

Component	Configuration
Framework	PyTorch / TensorFlow (ImageNet-pretrained backbones)
Compute	GPU-accelerated training environment
Image backbones	DenseNet121, ViT-B/16, EfficientNetB0 (all ImageNet-pretrained)
Metadata model	Dense MLP (64 $\rightarrow$ 128 $\rightarrow$ 256)
Tabular baseline	XGBoost

4.6 Evaluation Metrics

Every classifier in this report is evaluated on a common set of metrics, chosen specifically because accuracy alone is misleading on an imbalanced dataset (Section 1.4):

- **Accuracy** – overall proportion of correct predictions; reported for comparability with prior literature, but not treated as the primary metric given class imbalance.
- **Macro-F1** – the F1 score averaged uniformly across all classes rather than weighted by class frequency, which prevents a model from scoring well purely by favouring the majority class.
- **Macro/ROC-AUC** – the area under the ROC curve, averaged across classes, used as a threshold-independent measure of separability.
- **Sensitivity (recall) and specificity** – reported specifically for the binary melanoma/non-melanoma task in Chapter 6, since a missed melanoma (false negative) and a false alarm (false positive) carry very different real-world costs.

4.7 Hyperparameter Configuration

Table 4.4 lists the training hyperparameters used for the two primary baselines in Chapter 5, held identical across both models specifically so that the comparison in Section 5.4 isolates the effect of architecture rather than training configuration.

Table 4.4: Hyperparameter configuration for DenseNet121 and ViT-B/16 baselines.

Hyperparameter	Value
Backbone initialisation	ImageNet-pretrained
Epochs	20
Train/validation split	75/25
Input resolution	$224 \times 224 \times 3$
Metadata used	Age, sex, anatomical site
Dataset	MILK10k (ISIC 2025)

With the dataset, preprocessing, and training protocol now fixed, Chapter 5 applies this exact setup to build and compare the two baseline classifiers.

Chapter 5

Benchmark Model Implementation

Chapter 4 fixed the dataset and the experimental protocol. This chapter applies that protocol to build two independent single-modality baselines – DenseNet121 and ViT-B/16 – under identical training conditions on MILK10k, and reports a comparative analysis between them. This is the empirical counterpart to the theoretical CNN-versus-ViT comparison already laid out in Table 2.1.

5.1 Need for Benchmark Models

Before attempting any multimodal fusion architecture (Chapter 6) or interpretability layer (Chapter 7), it is necessary to establish how well the two underlying image architecture families perform on their own, under a shared, controlled protocol. Without this step, any later claim that fusion or metadata “helps” would have nothing fair to be measured against – a concern raised directly in Section 3.7 as a broader comparability gap in the published literature. DenseNet121 and ViT-B/16 were chosen specifically because they represent the two architecture families discussed throughout Chapter 2: a mature, parameter-efficient CNN, and a global-attention transformer, both initialised from ImageNet pretraining (Section 2.4).

5.2 DenseNet121 Baseline

Architecture overview. DenseNet121 [5] connects each layer to every other layer in a feed-forward fashion within a dense block, so that each layer receives the feature maps of all preceding layers as input. This encourages feature reuse and gives the network strong gradient flow during training, at a lower parameter cost than a comparably deep architecture without such connections – the theoretical property discussed generally in Section 2.3 and revisited here in its specific, published form.

[Insert Figure]

Description: DenseNet-121 architecture diagram showing dense blocks, transition layers, and the growth-rate connectivity pattern.

Suggested image: Original DenseNet architecture figure from Huang et al., “Densely Connected Convolutional Networks,” CVPR 2017

Source: arXiv:1608.06993 / official CVPR paper figure

Figure 5.1: DenseNet-121 architecture (placeholder).

Implementation. An ImageNet-pretrained DenseNet121 backbone was fine-tuned as an image-only classifier on MILK10k, following the transfer-learning recipe of Section 2.4, with age, sex, and anatomical site available in the dataset but not consumed by this particular baseline (it is deliberately image-only, to isolate the CNN architecture’s own performance before any metadata is introduced).

Training configuration. Table 5.1 lists the configuration used, matching Table 4.4.

Table 5.1: DenseNet121 training configuration.

Parameter	Value
Dataset	MILK10k (10,480 images, 11 classes)
Split	75/25 train-validation
Epochs	20
Initialisation	ImageNet-pretrained
Modality	Image-only

Results. Table 5.2 reports validation performance.

Table 5.2: DenseNet121 validation results on MILK10k.

Metric	Value
Accuracy	71.18%
Macro-F1	66.75%
Macro-AUC	0.9016

[Insert Figure]

Description: Confusion matrix for the DenseNet121 11-class MILK10k baseline.

Suggested image: Confusion matrix figure generated during this internship

Source: Internship presentation, Slide 5

Figure 5.2: DenseNet121 confusion matrix (11-class).

Observations. The model performs strongest on visually distinctive classes and struggles most on classes with overlapping visual presentation – melanoma versus melanocytic nevus (MEL vs NV), and basal cell carcinoma versus actinic keratosis (BCC vs AK) – consistent with the visual-similarity problem identified generally in Section 1.4 and with the CNN limitations reviewed in Section 3.2.

5.3 Vision Transformer Baseline

Architecture overview. ViT-B/16 [4] applies the patch-embedding, positional-encoding, and multi-head self-attention pipeline described in full in Section 2.5, using 16×16 patches and the “Base” configuration (12 encoder blocks). Its defining property relative to DenseNet121 is global receptive field from the first layer, as summarised theoretically in Table 2.1.

Implementation. ViT-B/16, initialised from ImageNet pretraining, was fine-tuned as an image-only classifier under the exact same protocol as DenseNet121 – same dataset, same split, same number of epochs – so that any performance difference between the two models can be attributed to architecture rather than training conditions.

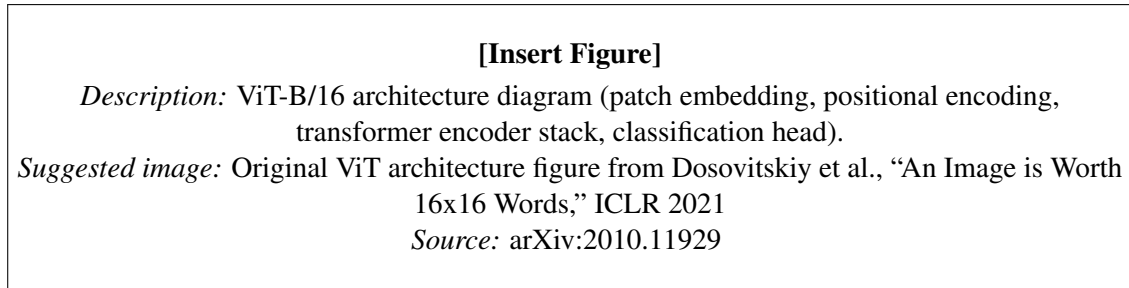


Figure 5.3: ViT-B/16 architecture (placeholder).

Training configuration. Table 5.3 lists the configuration, identical to Table 5.1 except for the backbone itself.

Table 5.3: ViT-B/16 training configuration.

Parameter	Value
Dataset	MILK10k (10,480 images, 11 classes)
Split	75/25 train-validation (same split as DenseNet121)
Epochs	20
Initialisation	ImageNet-pretrained
Modality	Image-only

Results. Table 5.4 reports validation performance.

Table 5.4: ViT-B/16 validation results on MILK10k.

Metric	Value
Accuracy	75.50%
Macro-F1	72.31%
Macro-AUC	0.9251

Observations. ViT-B/16 outperforms DenseNet121 on every metric under this identical protocol. The gain is most visible on classes where the diagnostically relevant cue is spread across the lesion rather than concentrated in one local region – exactly the kind of global relationship the self-attention mechanism (Section 2.5.3) is built to capture, and exactly the limitation of local convolution raised as motivation in Section 1.5.

5.4 Comparative Analysis

Table 5.5 places both single-modality baselines side by side, and Table 5.6 extends the comparison to Baseline 3, a CNN + gradient-boosted decision tree (GBDT) stacking pipeline built to test whether adding metadata through a fusion-of-predictions approach (Section 3.5) improves on either image-only baseline.

Baseline 3 follows the ConvNeXt-Base out-of-fold (OOF) probability plus XGBoost stacking pattern reviewed in Section 3.5, using a single backbone and a single GBDT rather than the multi-backbone, multi-GBDT ensembles used by top entries in the ISIC 2024 challenge. Two pipeline variants were evaluated: Pipeline 1 uses only the CNN’s OOF class probabilities plus metadata, and Pipeline 2 additionally engineers prediction-confidence features on top.

Table 5.5: DenseNet121 vs ViT-B/16 on MILK10k (11-class, identical protocol).

Metric	DenseNet121	ViT-B/16
Accuracy	71.18%	75.50%
Macro-F1	66.75%	72.31%
Macro/ROC-AUC	0.9016	0.9251
Validation set size	2,620 (25% split)	2,620 (25% split)
Modality	Image-only	Image-only

Table 5.6: Baseline 3 (CNN + GBDT stacking) results on MILK10k.

Metric	Pipeline 1	Pipeline 2
Accuracy	69.27%	68.99% (−0.29 pt)
Macro-F1 (11-class)	0.39	0.38
Log-loss	not computed	0.971
Feature count	46	51

Both Baseline 3 pipelines trail the simple image-only baselines already established in Table 5.5. This is a direct, empirical instance of the metadata-richness caveat raised in Section 3.5: only three metadata fields (age, sex, anatomical site) are available, and this small a set carries little signal beyond what the CNN’s own predicted probabilities already encode. Rare MILK10k classes with fewer than 20 training samples (Section 4.3) score close to zero F1 regardless of which pipeline is used, and a single backbone plus a single GBDT is simply not the same model as the multi-model ensembles used by the challenge winners this pattern is drawn from.

Table 5.7: Full ranking across all three benchmark models on MILK10k (11-class accuracy).

Rank	Model	Accuracy	Macro-F1	Modality
1	ViT-B/16	75.50%	72.31%	Image-only
2	DenseNet121	71.18%	66.75%	Image-only
3	Baseline 3 (CNN + GBDT)	69.27% (best)	0.39	Image + metadata (stacked)

The clearest finding of this chapter is a slightly uncomfortable one: on this dataset, under this protocol, adding metadata through prediction-level stacking did not help, and the simplest image-only ViT baseline remained the strongest model. This finding directly informs the design of the multimodal architecture reproduced in the next chapter – XAAI-Ledger fuses metadata at the representation level, through self-attention, rather than at the prediction level through stacking, which is a meaningfully different mechanism and is evaluated on its own terms there.

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Chapter 6

Implementation of XAAI-Ledger Framework

Chapter 5 closed on a slightly uncomfortable finding: prediction-level metadata stacking did not beat a simple image-only ViT baseline on MILK10k. This chapter turns to a different way of combining image and metadata – representation-level fusion through self-attention – by reproducing XAAI-Ledger, the multi-modal, explainable architecture reviewed in Section 3.6, directly from its published methodology.

6.1 Introduction

XAAI-Ledger was selected as the reproduction target for three reasons that follow directly from the research gap identified in Section 3.7. First, it fuses image and metadata at the representation level through multi-head self-attention rather than at the prediction level, making it a genuinely different mechanism from Baseline 3 (Section 5.4) and a fair second data point on whether metadata fusion helps on this problem. Second, it reports both Grad-CAM and SHAP explainability outputs as a core contribution, directly addressing the interpretability-that-stops-at-attribution gap discussed in Section 3.4. Third, its published methodology is detailed enough to attempt an independent reproduction, which is precisely the reproducibility gap Section 3.7 identified as under-served in the wider multimodal skin-cancer literature.

This chapter documents that reproduction as built – architecture, pipelines, training configuration, and results – and reports the outcome honestly, including where it falls short of the paper’s own numbers.

6.2 Overall Architecture

XAAI-Ledger is a dual-stream architecture: a frozen CNN image branch and an MLP metadata branch are each projected into a shared embedding space, concatenated, and passed through a multi-head self-attention fusion block before a final classification head. Figure 6.1 reproduces the overall workflow as implemented in this internship.

At a high level, the pipeline funnels data through two parallel streams – a frozen pretrained CNN for visual texture extraction and a dense MLP for clinical metadata – before projecting both into a shared transformer self-attention sequence layer. Table 6.1 summarises the end-to-end pipeline; the sections that follow expand each stage.

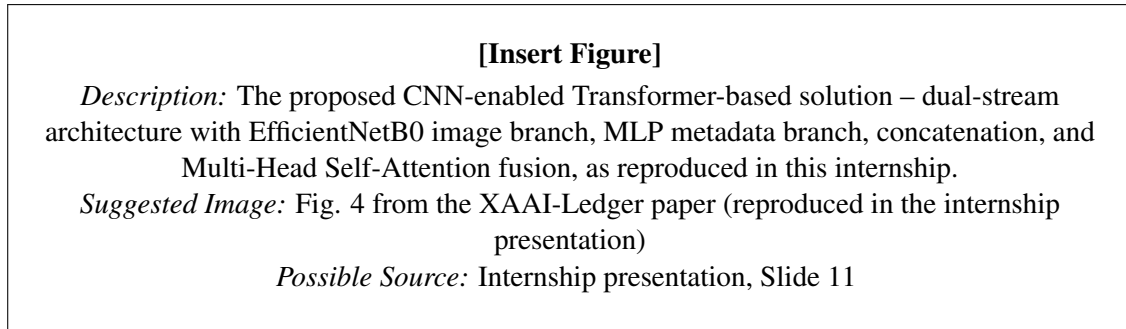


Figure 6.1: XAAI-Ledger overall architecture (as reproduced).

Table 6.1: XAAI-Ledger implementation pipeline, as reproduced in this internship.

Stage	Description
Image branch	Frozen EfficientNetB0, ImageNet-pretrained $\rightarrow$ 256-D visual feature vector
Metadata branch	Age, sex, lesion site (10-D, one-hot) $\rightarrow$ MLP (64 $\rightarrow$ 128 $\rightarrow$ 256) $\rightarrow$ 256-D vector
Fusion	Concatenate both 256-D vectors into a 512-D token, reshaped to a length-1 sequence
Attention	4-head Multi-Head Self-Attention, with Add & Norm and a Feed-Forward layer
Output head	Global Average Pooling + Dense sigmoid layer $\rightarrow$ Melanoma / Non-Melanoma

6.3 Image Processing Pipeline

The image branch follows the same $224 \times 224 \times 3$ preprocessing and normalisation steps established in Section 4.4, feeding into a frozen EfficientNetB0 [10] backbone. EfficientNet uses a compound scaling method that balances network depth, width, and input resolution under a fixed computational budget, which makes it an efficient choice for a frozen feature extractor relative to a larger, unscaled backbone. The backbone remains entirely frozen during training – only the downstream metadata MLP and fusion block are trained – so the $224 \times 224 \times 3$ dermoscopic image is converted into a fixed 256-dimensional visual embedding without any gradient updates flowing back into the CNN itself, consistent with the transfer-learning recipe of Section 2.4 in its feature-extraction (rather than fine-tuning) mode.

6.4 Metadata Processing Pipeline

The metadata branch consumes three fields – patient age, sex, and anatomical site of the lesion – following the same imputation and encoding strategy described in Section 4.4: median imputation for age, mode imputation for sex and site, one-hot encoding for the categorical fields, and MinMaxScaler normalisation for age. The resulting 10-dimensional metadata vector is passed through a small dense MLP (64 $\rightarrow$ 128 $\rightarrow$ 256), producing a 256-dimensional metadata embedding that is dimensionally matched to the image embedding from Section 6.3, so the two streams can be concatenated directly before fusion.

6.5 Feature Fusion Strategy

The two 256-dimensional embeddings are concatenated into a single 512-dimensional token and reshaped into a length-1 sequence, which is then passed through a 4-head Multi-Head Self-Attention block with residual Add & Norm and a feed-forward sub-layer – structurally the same encoder-block pattern introduced generally in Section 2.5.2, applied here to a fused image-plus-metadata token rather than a sequence of image patches.

Conceptually, self-attention lets the fused representation re-weight how much the final prediction relies on visual evidence versus clinical metadata, rather than fixing that balance in advance the way a simple concatenation-and-MLP fusion would. This is the representation-level counterpart to the prediction-level stacking used in Baseline 3 (Section 5.4), and is the main architectural distinction motivating this reproduction, as discussed in Section 3.6. Global Average Pooling followed by a single Dense sigmoid unit then converts the fused, attention-weighted representation into a binary Melanoma / Non-Melanoma prediction.

6.6 Explainability Module

Two explainability techniques, both reviewed generally in Section 3.4, were reproduced alongside the classifier itself:

- **Grad-CAM** [8] highlights the pixels in the dermoscopic image that most influenced the model’s prediction, computed from the gradients flowing into the last convolutional feature map of the EfficientNetB0 branch. The same original-image → heatmap → overlay sequence used in the source paper was reproduced, including a lesion bounding-box localisation step, run on a held-out validation image.
- **SHAP** estimates each clinical metadata feature’s contribution to the fused prediction by computing its Shapley value over the metadata branch, giving a feature-level explanation that complements Grad-CAM’s pixel-level one.

[Insert Figure]

Description: Grad-CAM lesion localisation – original image, heatmap, and overlay, for the paper’s output alongside this internship’s reproduced output.

Suggested Image: Grad-CAM comparison figure from the internship presentation

Possible Source: Internship presentation, Slide 13

Figure 6.2: Grad-CAM explainability – lesion localisation (paper vs. reproduced).

[Insert Figure]

Description: SHAP clinical metadata contribution – feature-level attribution for age, sex, and lesion site.

Suggested Image: SHAP attribution figure from the internship presentation

Possible Source: Internship presentation, Slide 14

Figure 6.3: SHAP explainability – clinical feature attribution.

The reproduced SHAP output found that patient age and lesion site (particularly lower extremity and head/neck) carry the highest average impact on the fused prediction, a pattern consistent with known clinical risk factors for melanoma and a reassuring sign that the metadata branch had learned something clinically sensible rather than an arbitrary correlation.

6.7 Training Procedure

Table 6.2 lists the training configuration used for the XAAI-Ledger reproduction, drawn directly from the internship presentation and notebook. Where a specific hyperparameter (optimizer, learning rate, batch size) was not explicitly recorded in the available materials, it is left as a placeholder rather than invented.

Table 6.2: XAAI-Ledger reproduction – training configuration.

Parameter	Value
Dataset	ISIC 2019 (25,331 samples)
Split	75/25 train-validation, stratified on class label
Validation set size	6,333 samples (5,202 non-melanoma, 1,131 melanoma)
Image branch	Frozen EfficientNetB0 (ImageNet-pretrained)
Metadata branch	Dense MLP (64 $\rightarrow$ 128 $\rightarrow$ 256)
Fusion	4-head Multi-Head Self-Attention
Optimizer	[Placeholder – not specified in available materials]
Learning rate	[Placeholder – not specified in available materials]
Batch size	[Placeholder – not specified in available materials]
Epochs	[Placeholder – not specified in available materials]
Loss function	Binary cross-entropy (sigmoid output)

6.8 Experimental Results

Table 6.3 compares the paper’s reported results against the numbers reproduced in this internship, both evaluated on the ISIC 2019 validation split described in Table 6.2.

Table 6.3: XAAI-Ledger – reproducibility assessment.

Metric	Paper (Reported)	Reproduced (Ours)
Accuracy	98.71%	74.00%
Sensitivity (Melanoma)	97.45%	16.00%
Specificity (Non-Melanoma Recall)	98.22%	~87.00%

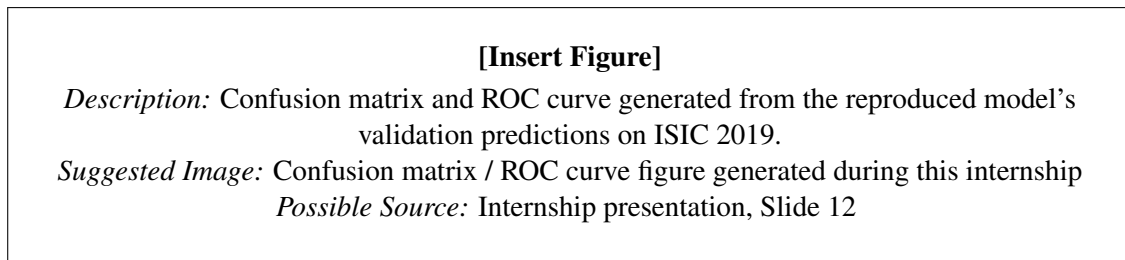


Figure 6.4: XAAI-Ledger reproduction – confusion matrix and ROC curve.

The headline result is a large gap on sensitivity specifically: the reproduced model catches only 16% of melanoma cases in the validation set, against a reported 97.45% in the paper, despite reasonable overall accuracy (74%) and non-melanoma recall ($\sim 87\%$). This pattern – high accuracy, high specificity, very low sensitivity – is the classic signature of a classifier that has learned to default toward the majority class under an unaddressed 4.6:1 class imbalance (Figure 1.4), exactly the leading candidate explanation flagged in Section 3.6: the paper's methodology section does not fully specify its class-imbalance handling strategy, and this reproduction's straightforward binary cross-entropy loss, without explicit class weighting or oversampling, evidently was not sufficient to compensate for the imbalance in the ISIC 2019 training split. Limited compute and time within the internship timeline also constrained the extent of hyperparameter and imbalance-handling exploration that could be attempted.

6.9 Chapter Summary

This chapter reproduced XAAI-Ledger's dual-stream, self-attention-fused architecture end-to-end – image branch, metadata branch, fusion block, and both explainability components – directly from its published methodology. The reproduction is architecturally faithful, and its explainability outputs (Grad-CAM localisation, SHAP metadata attribution) are qualitatively consistent with the original paper, but its classification sensitivity falls well short of the paper's reported numbers, most plausibly because of unaddressed class imbalance rather than an architectural shortfall. This honest gap, and the broader question of how to make a model's clinical reasoning traceable rather than opaque, motivated a further, more exploratory line of work carried out later in the internship, which the next chapter describes.

Chapter 7

Additional Development During the Internship

Chapter 6 closed with an honest reproduction gap and an open question: how should a model’s diagnostic reasoning be made traceable to clinically meaningful evidence, rather than left as an opaque probability score? After successfully implementing the XAAI-Ledger framework, additional development work was initiated during the later weeks of the internship (Table 1.1, Weeks 6–7) to explore this question further on the MILK10k dataset. This chapter documents that additional development honestly, as further exploration built on top of the reproduced framework – not as a novel, published architecture.

7.1 Motivation

The benchmark comparison in Section 5.4 and the XAAI-Ledger reproduction in Chapter 6 both point toward the same underlying tension: simple image-only baselines are strong and metadata fusion is difficult to get right, whether fused at the prediction level (Baseline 3) or the representation level (XAAI-Ledger). Rather than stopping at this observation, the remaining internship time was used to explore whether structuring the *prediction* itself – coarse-to-fine, and through clinically legible concepts – could offer a more interpretable path forward, without claiming to have solved the underlying fusion problem. The work described in this chapter should be read in that spirit: an exploratory extension building directly on the components already established in Chapters 5 and 6, evaluated honestly, including where it does not yet outperform the simpler baselines.

7.2 Representation Learning

The additional development reuses the same transfer-learning pattern established in Section 2.4 and applied throughout this report: a pretrained image backbone produces a fixed embedding, which downstream heads are trained on, rather than training a backbone from scratch. Image embeddings extracted from the MILK10k baselines (Chapter 5) and the CNN out-of-fold probabilities used in Baseline 3 (Section 5.4) form the representation dataset that the additional models in this chapter are trained on top of, rather than each model re-deriving its own features independently.

7.3 Disease Prediction

The disease-prediction component of this additional development, referred to as AlphaNet in the internship presentation, targets a coarser prediction than the 11-class MILK10k task used in Chapter 5: a binary malignant-versus-benign screening decision. Framing the problem at this coarser grain sidesteps some of the rare-class sparsity issues identified in Section 4.3 (several MILK10k classes have fewer than 20 training samples), and is closer to the actual triage decision a second-opinion screening tool (Section 1.1) would need to support in practice.

7.4 Clinical Concept Prediction

A separate component, referred to in the internship presentation as Baseline 4 (concept fusion), predicts a set of clinically meaningful attributes – rather than only a diagnosis label – as an intermediate step, extending the coarse-grained/fine-grained split described in Section 7.3 toward the full 11-class MILK10k label space. The intent mirrors the interpretability gap identified in Section 3.7: a diagnosis that passes through a clinically legible attribute first is more auditable than one that maps directly from pixels to a probability, even if, as the results in Section 7.8 make clear, this comes at a measurable accuracy cost at the full 11-class resolution.

7.5 Weighted CAAS

[Insert Figure]

Description: Workflow diagram of the Weighted CAAS (Confidence-Aware Aggregation Strategy) component used to combine per-class model confidence with concept-level predictions.

Suggested Image: Custom workflow diagram, to be sourced from the implementation notebook

Possible Source: Internal implementation notebook (not available in the materials used for this chapter)

Figure 7.1: Weighted CAAS workflow (placeholder).

Weighted CAAS is an internally-named confidence-weighting step used during this additional development to combine per-class prediction confidence with the concept-level signal described in Section 7.4, rather than treating every prediction as equally reliable regardless of the model’s own certainty. The exact weighting formulation used is documented in the internal implementation notebook rather than in the presentation materials available for this report; a precise description is therefore left as a placeholder above pending access to that notebook detail, in line with the instruction not to invent unavailable specifics. What can be stated with confidence, from the benchmark comparison in Section 5.4, is the purpose it was introduced to serve: Pipeline 2 in Baseline 3 (Section 5.4), which added prediction-confidence features on top of raw CNN probabilities, is the same general idea in its simplest form, and it did not improve on Pipeline 1 in that experiment – a result that directly motivated exploring a more deliberate, concept-aware confidence weighting in this later, more exploratory line of work.

7.6 Alpha–Beta Learning

The internship presentation names the disease-prediction and concept-prediction components introduced in Sections 7.3 and 7.4 as complementary stages – an **Alpha** stage (AlphaNet, coarse malignant-versus-benign screening) and a **Beta** stage (concept-level and fine-grained diagnosis prediction) – reflecting a coarse-to-fine learning structure rather than a single flat 11-class classifier. The Alpha stage’s role is to establish a reliable coarse triage signal first; the Beta stage’s role is to refine that signal into the clinically legible concepts and, ultimately, the full diagnostic label. This mirrors, at a modelling level, the same coarse-to-fine logic clinicians already use in practice: a first pass to flag concern, followed by a more detailed assessment.

7.7 Contextual Fusion

The final component of this additional development combines the Alpha-stage screening signal, the Beta-stage concept predictions, and the original clinical metadata (age, sex, anatomical site) into a single contextual representation before the final diagnostic decision – extending the fusion idea already used in Baseline 3 (prediction-level, Section 5.4) and XAAI-Ledger (representation-level, Chapter 6) to a three-way combination of screening signal, concept signal, and metadata.

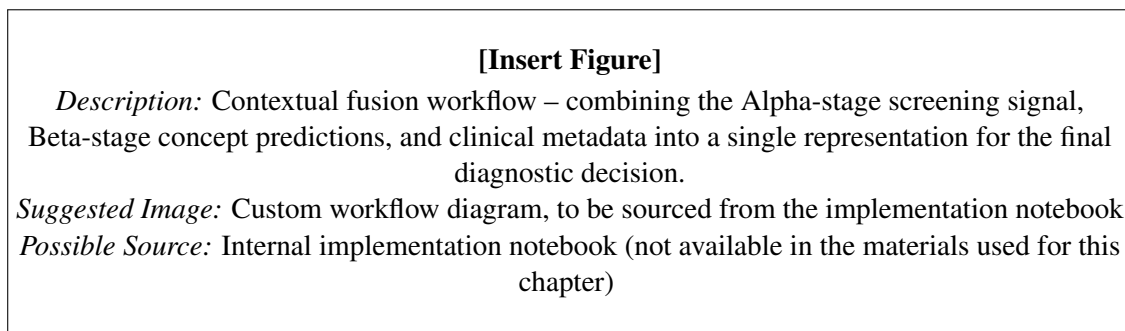


Figure 7.2: Contextual fusion workflow (placeholder).

7.8 Current Progress

Table 7.1 summarises what had been completed by the end of the internship against what remained ongoing, stated plainly so that unfinished work is not presented as complete.

The clearest completed result from this additional development is that concept-based calibration helps at the coarse-grained, binary level (AlphaNet, 0.845 AUC) but does not yet help at the full 11-class resolution, where Baseline 4 sits below the simple DenseNet121 and ViT-B/16 baselines already established in Chapter 5. This is discussed further, alongside the full model ranking, in Chapter 8.

7.9 Chapter Summary

This chapter documented the additional, exploratory development carried out after the XAAI-Ledger reproduction – a coarse-to-fine, concept-aware extension of the fusion ideas already established earlier in the report, evaluated honestly against the simpler baselines it builds on. With every model and pipeline in this report now described, Chapter 8 draws all of the results together into a single comparative analysis.

Table 7.1: Additional development – completed versus ongoing work.

Component	Status	Notes
AlphaNet (binary malignant/benign screening)	Completed	0.845 AUC on the binary sub-task
Baseline 4 (concept fusion, 11-class)	Completed	Sits below the simple image-only baselines (Table 5.7) at full 11-class resolution
Weighted CAAS confidence weighting	Ongoing	Formulation documented in the internal notebook only; not independently re-verified for this report
Alpha–Beta coarse-to-fine pipeline	Ongoing	Alpha stage validated; end-to-end Alpha→Beta hand-off not fully benchmarked
Contextual fusion (screening + concepts + metadata)	Ongoing	Workflow defined; full-pipeline results not yet available

Chapter 8

Results and Discussion

Chapters 5, 6, and 7 introduced, in turn, the single-modality baselines, the XAAI-Ledger reproduction, and the additional coarse-to-fine development. This chapter draws every result already reported in those chapters together into one comparative analysis, and interprets the patterns that emerge rather than restating individual numbers.

8.1 Evaluation Strategy

Every model in this report was evaluated using the metrics defined in Section 4.6 – accuracy, macro-F1, macro/ROC-AUC, and, for the binary melanoma task specifically, sensitivity and specificity – applied consistently so that results are comparable across chapters. The 11-class MILK10k models (Chapter 5, Section 7.8) are evaluated against each other directly, since they share an identical dataset and split; the binary ISIC 2019 XAAI-Ledger reproduction (Chapter 6) and the binary AlphaNet screening sub-task (Section 7.3) are evaluated on their own binary-task terms and are not directly comparable to the 11-class accuracy figures.

8.2 Benchmark Results

Table 8.1 restates the two single-modality baselines from Section 5.4 for reference.

Table 8.1: DenseNet121 vs ViT-B/16 on MILK10k (restated from Table 5.5).

Metric	DenseNet121	ViT-B/16
Accuracy	71.18%	75.50%
Macro-F1	66.75%	72.31%
Macro/ROC-AUC	0.9016	0.9251

ViT-B/16 outperformed DenseNet121 on every metric, and both comfortably outperformed the metadata-fusion attempts (Baseline 3, Baseline 4) at the full 11-class resolution, a pattern already established in Section 5.4 and re-examined below in Section 8.5.

8.3 XAAI-Ledger Results

The XAAI-Ledger reproduction (Table 6.3) matched the paper’s overall accuracy reasonably closely (74.00% versus a reported 98.71%, though accuracy alone is a misleading headline number here) but fell far short on melanoma sensitivity (16.00% versus a reported 97.45%), for the class-imbalance reasons discussed in Section 6.8. The reproduction’s explainability outputs – Grad-CAM localisation and SHAP metadata attribution – were qualitatively consistent with the paper despite the sensitivity gap, indicating that the fusion architecture itself was faithfully reproduced even though its training did not fully compensate for class imbalance.

8.4 Additional Development Results

Table 8.2 reports the additional-development results that are actually available, per Table 7.1; components still marked Ongoing in Section 7.8 are left as placeholders here rather than reported.

Table 8.2: Additional development – results actually available.

Component	Metric	Value
AlphaNet (binary malignant/benign)	AUC	0.845
Baseline 4 (concept fusion, 11-class)	Accuracy	Below Table 8.1 baselines (Table 5.7)
Weighted CAAS	–	[Placeholder – ongoing, Section 7.8]
Contextual fusion (full pipeline)	–	[Placeholder – ongoing, Section 7.8]

8.5 Comparative Analysis

Table 8.3 places every model with a completed 11-class or binary result side by side, extending Table 5.7 with the binary-task results from Chapters 6 and 7 shown separately, since they are not on the same label space as the 11-class MILK10k models.

Table 8.3: Comparative summary across all models evaluated in this report.

Model	Task	Accuracy	AUC	Modality
ViT-B/16	MILK10k, 11-class	75.50%	0.9251	Image-only
DenseNet121	MILK10k, 11-class	71.18%	0.9016	Image-only
Baseline 3 (CNN + GBDT)	MILK10k, 11-class	69.27%	not computed	Image + metadata (stacked)
Baseline 4 (concept fusion)	MILK10k, 11-class	below above baselines	not computed	Image + metadata + concepts
AlphaNet	Binary (malignant/benign)	–	0.845	Image + metadata
XAAI-Ledger (reproduced)	ISIC 2019, binary	74.00%	not computed	Image + metadata (attention fusion)

Two patterns stand out consistently across this table. First, at the full, fine-grained diagnostic resolution (11-class MILK10k), every metadata-fusion attempt – prediction-level (Baseline 3) and concept-level (Baseline 4) alike – sits below the simple image-only baselines. Second, at a coarser, binary resolution, added structure does pay off: AlphaNet’s 0.845 AUC on malignant-versus-benign screening is a genuinely useful result, and it is achieved with exactly the kind of coarse-to-fine framing described in Section 7.6.

8.6 Error Analysis

The DenseNet121 and ViT-B/16 baselines both struggle most on visually overlapping classes – MEL versus NV, BCC versus AK (Section 5.2) – consistent with the visual-similarity problem raised in Section 1.4. The metadata-fusion baselines fail differently: Baseline 3 and Baseline 4 both score close to zero F1 on MILK10k classes with fewer than 20 training samples (Section 4.3), regardless of pipeline, because a handful of metadata fields and a single backbone simply cannot compensate for a class the model has barely seen. The XAAI-Ledger reproduction’s failure mode is different again – not visual confusion, but a systematic bias toward the majority (non-melanoma) class under untreated 4.6:1 imbalance, reflected directly in its 16% melanoma sensitivity (Section 6.8).

Taken together, these are three distinct failure modes with three distinct causes: visual ambiguity (baselines), data sparsity (fusion baselines), and class imbalance (XAAI-Ledger reproduction) – and each points toward a different, specific fix rather than a single blanket “get more data” recommendation: targeted augmentation or fine-grained feature engineering for the first, oversampling or synthetic minority generation for rare MILK10k classes for the second, and explicit class weighting or focal loss for the third.

8.7 Discussion

The recurring theme across Chapters 5–7 is that architectural sophistication and metadata availability do not automatically translate into better performance – they help only when the extra signal is genuinely informative and there is enough of it, echoing the caveat raised in Section 3.5. ViT-B/16’s edge over DenseNet121 (Section 5.3) is best explained by its global receptive field being better suited to lesion-wide cues than DenseNet’s local convolutional filters (Section 2.5.3), a difference that shows up clearly on classes where the diagnostic cue is spread across the lesion rather than concentrated locally.

Why did fusion under-perform at fine-grained resolution but help at coarse resolution? The most likely explanation is signal-to-noise: with only three metadata fields and eleven diagnostic classes, the metadata’s marginal information is spread thin across many decision boundaries, most of which it has little to say about; collapsed to a single binary decision (malignant versus benign), that same metadata has far fewer boundaries to help with and therefore contributes more per boundary – consistent with AlphaNet’s stronger binary AUC (0.845) against the weaker full-resolution Baseline 4 result. This reading, rather than treating fusion as simply “not working,” is what motivates the future-scope discussion in Chapter 9: the right question is not whether to fuse metadata, but at what level of prediction granularity fusion is worth the added complexity.

Chapter 9

Conclusion and Future Scope

9.1 Internship Summary

This report has followed a single seven-week arc from first principles to an honestly evaluated reproduction and a further exploratory extension. Figure 9.1 restates that arc as it actually unfolded: foundational computer vision (Chapter 2) built the vocabulary needed to understand the Vision Transformer (Section 2.5), which in turn made it possible to build and fairly compare two independent benchmark models – DenseNet121 and ViT-B/16 (Chapter 5) – under an identical protocol. That comparison then set the baseline against which a published multi-modal, explainable architecture, XAAI-Ledger, was reproduced end-to-end (Chapter 6), and the honest shortfall observed in that reproduction directly motivated the additional, coarse-to-fine development explored in the final weeks of the internship (Chapter 7).

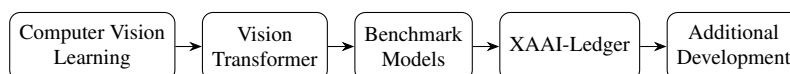


Figure 9.1: The internship arc summarised: from foundational computer vision through to the additional coarse-to-fine development.

9.2 Key Learnings

Table 9.1 summarises the key learnings from the internship across a few broad categories, drawn directly from the work documented in the preceding chapters rather than as a generic list.

9.3 Limitations

Several limitations apply across the work in this report, stated plainly rather than as self-criticism:

- **Compute and time.** All experiments were conducted within a seven-week internship timeline (Table 1.1) and the computational resources available to that timeline, which constrained the extent of hyperparameter search and class-imbalance handling that could be attempted, particularly for the XAAI-Ledger reproduction (Section 6.8).
- **Metadata richness.** Only three coarse metadata fields (age, sex, anatomical site) were available across every dataset used, which Section 8.7 identifies as the most likely explanation for fusion under-performing at fine-grained resolution.

Table 9.1: Key learnings from the internship.

Category	Learning
Technical skills	Implementing and fine-tuning CNN and ViT backbones, and building a multi-modal attention-fusion architecture from a paper’s methodology alone (Chapter 6).
Research skills	Reading a published architecture closely enough to reproduce it independently, and reporting a reproduction gap honestly rather than adjusting the narrative to fit expectations (Section 6.8).
Programming skills	Building consistent, comparable training and evaluation pipelines across several architectures under a fixed protocol (Section 4.7).
Medical imaging	Understanding class imbalance, metadata relevance, and failure modes specific to dermoscopic image classification (Section 8.6).
Deep learning	Direct, first-hand experience with transfer learning, self-attention, and representation- versus prediction-level fusion strategies (Chapters 2, 5–7).
Professional growth	Framing exploratory, unfinished work honestly as “additional development” rather than overstating its completeness or novelty (Chapter 7).

- **Unfinished components.** As stated plainly in Table 7.1, Weighted CAAS, the full Alpha–Beta hand-off, and the contextual fusion pipeline remain ongoing rather than completed, and their eventual results cannot yet be reported.
- **Single-institution, retrospective data.** All datasets used (Table 4.1) are public research benchmarks rather than prospectively collected clinical data, so none of the results in this report constitute evidence of real-world clinical performance.

9.4 Future Scope

The most direct next step, already sketched during the internship and reproduced here from the internship presentation, is a Concept Bottleneck Model (CBM) built on top of the paired clinical-and-dermoscopic MILK10k dataset (Section 4.2): a fusion encoder first predicts a set of human-defined clinical concepts – pigmentation, ulceration, erythema, vessel pattern, asymmetry, and border – and only then maps those concepts, through an interpretable linear classifier, to a diagnosis across the full 48 fine-grained ISIC-DX categories. Figure 9.2 reproduces this proposed pipeline.

Directly building on this proposal, and grounded in the specific weaknesses observed across Chapters 6–8, the following directions represent the most promising next steps:

- **Concept Bottleneck Models (CBMs).** As above – trading a black-box probability score for a diagnosis that is traceable to dermatologist-legible evidence, and that a clinician can correct at inference time by overriding a wrong concept prediction.
- **Clinical explainability.** Extending the Grad-CAM and SHAP outputs already reproduced in Chapter 6 to the concept level, so that a concept prediction itself – not just the final diagnosis – can be explained and audited.

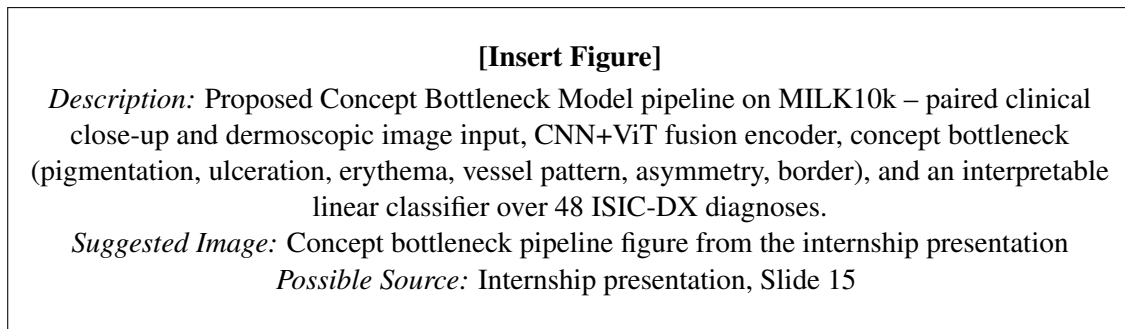


Figure 9.2: Proposed Concept Bottleneck Model pipeline (future work).

- **Generalization.** Testing whether models trained on one dataset (e.g. ISIC 2019) hold up on another (HAM10000, MILK10k), directly addressing the limited-and-inconsistent-data problem raised in Section 1.4.
- **External validation.** Evaluating any future model on data from institutions or imaging devices not represented in the training datasets used in this report, before any claim of clinical readiness could be entertained.
- **Foundation models.** Exploring larger, more general-purpose pretrained vision-language or medical-imaging foundation models as an alternative to the fixed ImageNet-pretrained backbones used throughout this report, particularly for the concept-prediction stage of a CBM.
- **Clinical decision support integration.** Framing any future model explicitly as a second-opinion tool (Section 1.1) integrated into a clinician’s existing workflow, rather than as a standalone predictive system.
- **Model deployment.** Moving beyond the notebook-based experimentation used throughout this internship toward a reproducible, versioned deployment pipeline, a prerequisite for any of the above to be clinically useful rather than only academically interesting.

9.5 Final Conclusion

This internship set out to understand, implement, and critically evaluate the progression from classical computer vision to CNNs, to ViTs, to multi-modal, explainable architectures for skin cancer detection – and, on its own terms, it did exactly that. Two independently trained benchmark classifiers were compared fairly under an identical protocol; a published multi-modal, explainable architecture was reproduced end-to-end from its methodology alone, with its shortfall reported honestly rather than smoothed over; and a further, coarse-to-fine exploratory extension was built and evaluated on top of that reproduction, with completed and ongoing components kept clearly distinct throughout. The results were not uniformly flattering – reproduced sensitivity fell well short of the source paper, and metadata fusion did not help at fine-grained resolution – but that honesty is, itself, the more useful outcome of a seven-week internship than an inflated headline number would have been. The concept-bottleneck direction proposed in Section 9.4 gives a concrete, evidence-grounded path for whoever continues this work next: not a bigger model, but a more legible one.